

MAGMACORE

- Magma Core is an attempt to implement a High Quality Data Modelling (HQDM) framework based on the model presented in Dr Matthew West's book "Developing High Quality Data Models"
- You would be well advised to read Dr. West's book because I might not be explaining all the details here.
- It has been used to develop a number of prototype systems, one of which is still in use, and was first published by GCHQ in the UK
- My fork is available on GitHub: <https://github.com/twalmsley/MagmaCore>
- The original is here: <https://github.com/gchq/MagmaCore>
- The original appears to have fallen out of maintenance, hence my fork. I wrote pretty much all of the code, so I will try to maintain it and develop it further in my fork.

BACKGROUND DETAILS

- MagmaCore is written in Java.
- It was published by GCHQ from an internal project, and is intended to foster development of HQDM-based systems in the hope of improving data modelling and data model compatibility to make data sharing more straightforward.
- It is based on RDF and the Apache Jena library for storing data as an RDF graph.
- Apache Jena supports inferencing rule which can be used to enhance the underlying model with additional facts not directly present in the original dataset.

ABOUT THE AUTHOR

- My name is Tony Walmsley, I am a software engineer with over 35 years experience in a wide variety of commercial software systems.
- I am the original developer of MagmaCore, and plan to continue its development and maintenance in my spare time as a freelance software engineer.
- I am based in Swindon in the UK.

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- Uses Java 17 and is built using Maven, which results in the following built libraries:
 - **core** - Database and Database Access Services
 - **hqdm** - The HQDM Top Level Ontology implementation
 - **hqdm-canonical** - Classes for using HQDM in a particular canonical way
 - **model-extension-example** - An example of how to add domain-specific extensions
 - **examples** - Various examples of how to use the library.

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CORE PACKAGE

- Use for connecting to a local or remote Jena database, or a SPARQL endpoint.
- Uses a `MagmaCoreServiceFactory` to create a `MagmaCoreService` instance for interacting with the selected database.
- Contains utilities for batch CRUD operations and for creating random UUIDs.

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HQDM PACKAGE

- Contains Service classes for creating HQDM model instances and relationships.
- You can use the services or create the model manually.
- Contains an extension mechanism for automatically discovering extension modules that contain domain-specific extensions to the core Top Level Ontology.
- The `HqdmObjectFactory` helps when creating HQDM entities from RDF Triples.

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HQDM-CANONICAL PACKAGE

- Contains a set of `*Builder` classes to make creation of the HQDM data model more straightforward.
- The `*Builder` classes are designed to indicate what relationships to other objects are needed, and which are considered mandatory by the original HQDM data model.
- The `*Builder` methods can be chained together before the final call to the `build()` method validates the object and returns the new instance or throws an exception to indicate the error.

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MODEL-EXTENSION-EXAMPLE

- Implements an example of an SPI (Service Provider Interface) extension module for HQDM.
- The example adds a few simple domain classes that extend existing HQDM package classes, and which can be used as if they were provided by the MagmaCore library.
- This example doesn't use `*Builder` classes, but these could be provided by the extension if preferred.

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EXAMPLES PACKAGE

- There are several packages within this package:
 - **data** - Creates some objects that are used by multiple examples
 - **extensions** - Shows how to use the example extension module described above
 - **functional** - Some examples that use a functional-programming style.
 - **service** - Examples of how to use the core `MagmaCoreService` class.
 - **signs** - Examples for how to create signs.

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HOW TO BUILD LOCALLY

```
git clone git@github.com:twalmsley/MagmaCore.git  
mvn clean install
```

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FURTHER INFORMATION

- See the README
- <https://twalmsley.github.io/blog1/blog1.html>
- https://hqdmtop.github.io/hqdmFramework/hqdm_framework/hqdm_framework.htm#entity
- <https://github.com/hqdmTop/hqdmFramework>
- <https://github.com/hqdmTop/hqdmOntology>
- <mailto:tony@aosd.co.uk>

